

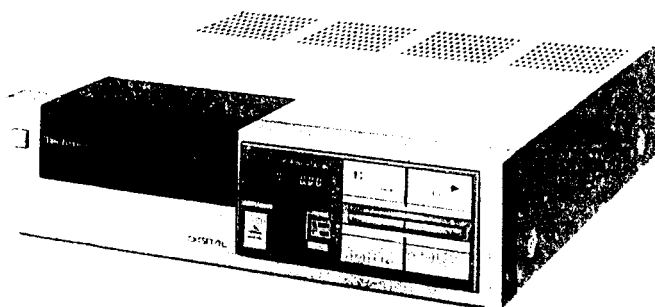
# Service Manual

**DIGITAL**

Compact Disc Player

**SL-P7/(K)**

[E], [EK], [XL], [EG],  
[EB], [EH], [EF], [Ei],  
[XA], [PA], [PE], [PC]



\*The colors of this model include silver and black.  
\*The black type model is provided with (K) in the Service Manual.

## Areas

- \* [E] is available in Switzerland and Scandinavia.
- \* [EK] is available in United Kingdom.
- \* [XL] is available in Australia.
- \* [EG] is available in F.R Germany.
- \* [EB] is available in Belgium.
- \* [EH] is available in Holland.
- \* [EF] is available in France.
- \* [Ei] is available in Italy.
- \* [XA] is available in Southeast Asia, Oceania, Africa, Middle Near East and Central South America.
- \* [PA] is available in far East PX.
- \* [PE] is available in European Military.
- \* [PC] is available in European Audio Club.

## Specifications

Specifications are subject to change without notice for further improvement.  
Weight and dimensions shown are approximate.

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>■ <b>Audio</b> <ul style="list-style-type: none"> <li>No. of channels: 2 (left and right stereo)</li> <li>Frequency response: 2 ~ 20,000 Hz <math>\pm</math> 0.5 dB</li> <li>Dynamic range: more than 96 dB</li> <li>S/N ratio: more than 96 dB</li> <li>Total harm. dist.: less than 0.003% (1 kHz, 0 dB)</li> <li>Channel separation: more than 90 dB</li> <li>Wow and flutter: Below measurable limit</li> </ul> </li> <li>■ <b>Signal Format</b> <ul style="list-style-type: none"> <li>Sampling frequency: 44.1 kHz</li> <li>Correction system: Technics Super Decoding Algorithm</li> </ul> </li> <li>■ <b>Pickup</b> <ul style="list-style-type: none"> <li>Type: Astigma 3-beam</li> <li>Light source: Semiconductor laser</li> <li>Wavelength: 800 nm</li> </ul> </li> <li>■ <b>General</b> <ul style="list-style-type: none"> <li>Power supply: ~110-120/220-240V, 50 or 60 Hz</li> <li>Power consumption: 30W</li> <li>Output voltage: 2.0 volts (at 0 dB)</li> <li>Output impedance: 330 ohms</li> <li>Load impedance: more than 5 kohms</li> <li>Dimensions (WxDxH): 31.5 x 32.5 x 8.8cm<br/>When disc holder is opened (31.5x45.1x8.8cm)</li> <li>Weight: 4.9 kg</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>■ <b>Functions</b> <ul style="list-style-type: none"> <li>Search modes: Manual search, Index search<br/>Skip search</li> <li>Display functions: All tracks, All playing time,<br/>Playing track, Playing time<br/>Index number</li> <li>Operation buttons: Basic buttons; 7<br/>Repeat button; 1</li> <li>Disc loading: Motor driven, lateral loading</li> </ul> </li> <li>■ <b>Disc specifications</b> <ul style="list-style-type: none"> <li>Diameter: 12 cm</li> <li>Thickness: 0.12 cm</li> <li>Spindle hole: 1.5 cm</li> <li>Min. inside diameter of recorded section: 5 cm</li> <li>Max. outside diameter of recorded section: 11.6 cm<br/>(disc is played from the inside to the outside edge)</li> <li>Direction of rotation: Counterclockwise<br/>(seen from recorded side of disc)</li> <li>Tracking speed: 1.2 to 1.4 meters/second, CLV<br/>(Constant Linear Velocity)</li> <li>Rotations per minute: About 500 to 200 rpm</li> <li>Playing time: About 60 minutes (up to 75 min. can be contained)</li> <li>Track pitch: 1.6<math>\mu</math>m</li> <li>Material: Clear plastic</li> </ul> </li> </ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Technics

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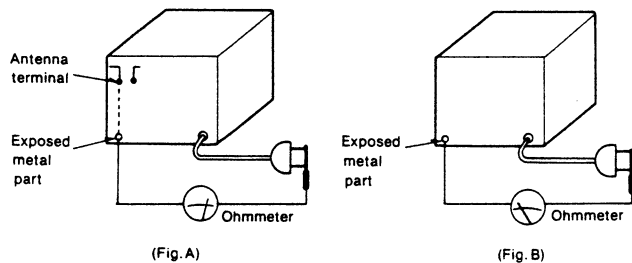
## SAFETY PRECAUTION

- Before servicing, unplug the power cord to prevent an electric shock.
- When replacing parts, use only manufacturer's recommended components for safety.
- Check the condition of the power cord. Replace if wear or damage is evident.
- After servicing, be sure to restore the lead dress, insulation barriers, insulation papers, shields, etc.
- Before returning the serviced equipment to the customer, be sure to make the following insulation resistance test to prevent the customer from being exposed to a shock hazard.

### INSULATION RESISTANCE TEST

- Unplug the power cord and short the two prongs of the plug with a jumper wire.
- Turn on the power switch.
- Measure the resistance value with ohmmeter between the jumpered AC plug and each exposed metal cabinet part, such as screwheads antenna, control shafts, handle brackets, etc. Equipment with antenna terminals should read between  $3M\Omega$  and  $5.2M\Omega$  to all exposed parts. (Fig. A) Equipment without antenna terminals should read approximately infinity to all exposed parts. (Fig. B)

**Note:** Some exposed parts may be isolated from the chassis by design. These will read infinity.

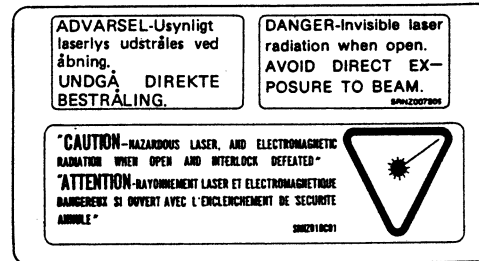
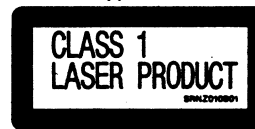


Resistance =  $3M\Omega$ — $5.2M\Omega$

Resistance = Approx  $\infty$

- If the measurement is outside the specified limits, there is a possibility of a shock hazard. The equipment should be repaired and rechecked before it is returned to the customer.

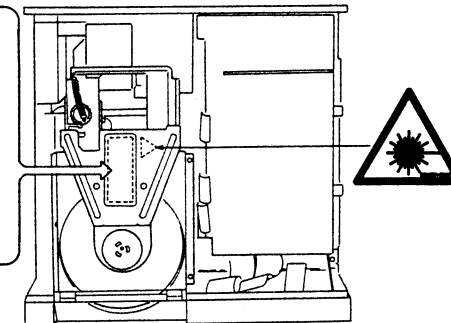
Caution : This product utilizes a laser diode.  
ADVASEL : I dette apparat anvendes laser.



### Use of caution labels

|            | U.S.A. | Canada | Europe | Others |
|------------|--------|--------|--------|--------|
| SRNZ010S01 | X      | X      | ○      | ○      |
| SRNZ007S05 | ○      | X      | ○      | ○      |
| SRNZ010C01 | X      | ○      | X      | X      |
| SRNZ010S02 | X      | X      | ○      | ○      |

Note: ○ Mark Label is used. X Mark Label is not used.



## CHANGES

### REPLACEMENT PARTS LIST

#### Notes:

- Mentioned in this parts list are only those changed in Model No. SL-P7 for destination [E] area.
- Important safety notice: Components identified by  $\Delta$  mark have special characteristics important for safety. When replacing any of these components, use only manufacturer's specified parts.
- The "S" mark is service standard parts and may differ from production parts.

| Ref. No.            | Change of Part No. |                    | Part Name & Description        | Per Set (Pcs.) | Remarks |
|---------------------|--------------------|--------------------|--------------------------------|----------------|---------|
|                     | SL-P7/(K)<br>(OLD) | SL-P7/(K)<br>(NEW) |                                |                |         |
| INTEGRATED CIRCUITS |                    |                    |                                |                |         |
| IC3                 | AN7915             | AN7915T            | Regulator                      | 1              |         |
| IC403               | MN4053B            | MN4053BS           | Analog Switch                  | 1              |         |
| IC805 (A)           | SVIUPD4053BC       | SVILM833NA         | Operational Amplifier          | 1              |         |
| IC805 (B)           |                    | SVIUPD4053BC       | Analog Switch                  | 1              |         |
| IC861               | SVILM833NA         | Deletion           | -----                          | 0              |         |
| TRANSISTOR          |                    |                    |                                |                |         |
| Q208                | 2SC1344            | 2SC1328-T          | CLV Control                    | 1              | ⑤       |
| DIODES              |                    |                    |                                |                |         |
| D206                | Addition           | MA165              | Traverse Servo Control         | 1              |         |
| D207                | MA165              | Deletion           | -----                          | 0              |         |
| D403                | MA165              | Deletion           | -----                          | 0              |         |
| VARIABLE RESISTOR   |                    |                    |                                |                |         |
| VR104               | EVN32CA00B14       | EVN32CA00B53       | Tracking Balance Adj., 5kΩ (B) | 1              |         |

| Ref. No.           | Change of Part No. |                    | Part Name & Description   | Per Set<br>(Pcs.) | Remarks |
|--------------------|--------------------|--------------------|---------------------------|-------------------|---------|
|                    | SL-P7/(K)<br>(OLD) | SL-P7/(K)<br>(NEW) |                           |                   |         |
| VARIABLE CAPACITOR |                    |                    |                           |                   |         |
| CT401              | ECV1ZW25X53N       | ECV1ZW25X53T       | Trimmer, PLL Adj.         | 1                 |         |
| RESISTORS          |                    |                    |                           |                   |         |
| R112               | ERDS2TJ101         | ERDS2TJ221         | Carbon, 1/4W, 220Ω, ± 5%  | 1                 |         |
| R122               | ERDS2TJ334         | ERDS2TJ184         | Carbon, 1/4W, 180kΩ, ± 5% | 1                 |         |
| R125               | ERDS2TJ183         | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R127               | ERDS2TJ562         | ERDS2TJ822         | Carbon, 1/4W, 8.2kΩ, ± 5% | 1                 |         |
| R133               | ERDS2TJ392         | ERDS2TJ222         | Carbon, 1/4W, 2.2kΩ, ± 5% | 1                 |         |
| R141               | ERDS2TJ102         | ERDS2TJ101         | Carbon, 1/4W, 100Ω, ± 5%  | 1                 |         |
| R143               | ERDS2TJ562         | ERDS2TJ822         | Carbon, 1/4W, 8.2kΩ, ± 5% | 1                 |         |
| R247               | ERDS2TJ104         | ERDS2TJ273         | Carbon, 1/4W, 27kΩ, ± 5%  | 1                 |         |
| R267               | ERDS2TJ103         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R268               | ERDS2TJ103         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R273               | Addition           | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R274               | Addition           | ERDS2TJ273         | Carbon, 1/4W, 27kΩ, ± 5%  | 1                 |         |
| R275               | Addition           | ERDS2TJ683         | Carbon, 1/4W, 68kΩ, ± 5%  | 1                 |         |
| R276               | Addition           | ERDS2TJ102         | Carbon, 1/4W, 1kΩ, ± 5%   | 1                 |         |
| R319               | ERDS2TJ560         | ERD25FJ560         | Carbon, 1/4W, 56Ω, ± 5%   | 1                 | Ⓢ ⚠     |
| R409               | ERDS2TJ223         | ERDS2TJ273         | Carbon, 1/4W, 27kΩ, ± 5%  | 1                 |         |
| R411               | ERD25FJ103         | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R412               | ERD25FJ152         | ERDS2TJ152         | Carbon, 1/4W, 1.5kΩ, ± 5% | 1                 |         |
| R413               | ERDS2TJ273         | ERDS2TJ223         | Carbon, 1/4W, 22kΩ, ± 5%  | 1                 |         |
| R423               | ERD25TJ104         | ERDS2TJ104         | Carbon, 1/4W, 100kΩ, ± 5% | 1                 |         |
| R424               | ERD25FJ102         | Deletion           | -----                     | 0                 |         |
| R425               | ERD25FJ103         | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R426               | ERD25FJ561         | ERDS2TJ561         | Carbon, 1/4W, 560Ω, ± 5%  | 1                 |         |
| R427               | ERD25FJ682         | ERDS2TJ682         | Carbon, 1/4W, 6.8kΩ, ± 5% | 1                 |         |
| R428               | ERD25FJ471         | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R429               | ERD25FJ103         | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R430               | ERD25FJ221         | ERDS2TJ221         | Carbon, 1/4W, 220Ω, ± 5%  | 1                 |         |
| R553               | ERDS2TJ153         | ERDS2TJ123         | Carbon, 1/4W, 12kΩ, ± 5%  | 1                 |         |
| R582               | Addition           | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R589               | ERDS2TJ103         | Deletion           | -----                     | 0                 |         |
| R610               | ERDS2TJ333         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R619               | ERDS2TJ333         | Deletion           | -----                     | 0                 |         |
| R621               | Addition           | ERDS2TJ333         | Carbon, 1/4W, 33kΩ, ± 5%  | 1                 |         |
| R622               | Addition           | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R623               | Addition           | ERDS2TJ103         | Carbon, 1/4W, 10kΩ, ± 5%  | 1                 |         |
| R851               | ERD25TJ183         | ERDS2TJ183         | Carbon, 1/4W, 18kΩ, ± 5%  | 1                 |         |
| R852               | ERD25FJ102         | ERDS2TJ102         | Carbon, 1/4W, 1kΩ, ± 5%   | 1                 |         |
| R853               | ERD25FJ682         | ERDS2TJ682         | Carbon, 1/4W, 6.8kΩ, ± 5% | 1                 |         |
| R854               | ERD25FJ472         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R855               | ERD25FJ472         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R856               | ERD25FJ682         | ERDS2TJ682         | Carbon, 1/4W, 6.8kΩ, ± 5% | 1                 |         |
| R857               | ERD25FJ102         | ERDS2TJ102         | Carbon, 1/4W, 1kΩ, ± 5%   | 1                 |         |
| R858               | ERD25FJ471         | ERDS2TJ471         | Carbon, 1/4W, 470Ω, ± 5%  | 1                 |         |
| R859               | ERD25FJ822         | ERDS2TJ822         | Carbon, 1/4W, 8.2kΩ, ± 5% | 1                 |         |
| R863               | ERD25FJ472         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |
| R864               | ERD25FJ682         | ERDS2TJ682         | Carbon, 1/4W, 6.8kΩ, ± 5% | 1                 |         |
| R865               | ERD25FJ472         | ERDS2TJ472         | Carbon, 1/4W, 4.7kΩ, ± 5% | 1                 |         |

| Ref. No.   | Change of Part No. |                      | Part Name & Description             | Per Set<br>(Pcs.) | Remarks |
|------------|--------------------|----------------------|-------------------------------------|-------------------|---------|
|            | SL-P7/(K)<br>(OLD) | → SL-P7/(K)<br>(NEW) |                                     |                   |         |
| CAPACITORS |                    |                      |                                     |                   |         |
| C23        | ECEA1HU100         | Deletion             | -----                               | 0                 |         |
| C26        | ECEA1HU100         | Deletion             | -----                               | 0                 |         |
| C94        | Addition           | ECEA50Z1             | Electrolytic, 50V, 1μF              | 1                 | Ⓢ       |
| C95        | Addition           | ECEA1HU100           | Electrolytic, 50V, 10μF             | 1                 | Ⓢ       |
| C96        | Addition           | ECEA1HU100           | Electrolytic, 50V, 10μF             | 1                 | Ⓢ       |
| C119       | ECKD1H472KB        | ECFR1E104ZF          | Ceramic, 25V, 0.1μF, +80%<br>-20%   | 1                 |         |
| C126       | ECKD1H152KB        | ECQM1H562KV          | Polyester, 50V, 0.0056μF, ± 10%     | 1                 |         |
| C127       | ECQM1H103KV        | ECQM1H223KV          | Polyester, 50V, 0.022μF, ± 10%      | 1                 |         |
| C133       | ECKD1H681KB        | ECQM1H102KV          | Polyester, 50V, 0.001μF, ± 10%      | 1                 |         |
| C134       | ECKD1H681KB        | ECCD1H681K           | Ceramic, 50V, 680pF, ± 10%          | 1                 |         |
| C135       | ECKD1H473ZF        | ECFR1E104ZF          | Ceramic, 25V, 0.1μF, +80%<br>-20%   | 1                 |         |
| C205       | ECEA1HN010         | ECEA1HN010S          | Electrolytic, 50V, 1μF              | 1                 | Ⓢ ⚠     |
| C225       | ECEA50Z2R2         | ECEA50Z3R3           | Electrolytic, 50V, 0.33μF           | 1                 | Ⓢ       |
| C234       | Addition           | ECEA1CN100S          | Electrolytic, 16V, 10μF             | 1                 | Ⓢ ⚠     |
| C306       | ECEAQN220S         | ECEA1AN220S          | Electrolytic, 10V, 22μF             | 1                 | ⚠       |
| C503       | ECEA125N4R7        | ECEA25N4R7           | Electrolytic, 25V, 4.7μF            | 1                 | Ⓢ ⚠     |
| C504       | ECEA125N4R7        | ECEA25N4R7           | Electrolytic, 25V, 4.7μF            | 1                 | Ⓢ ⚠     |
| C506       | ECUV1E102JC        | ECKD1H102KB          | Ceramic, 50V, 0.001μF, ± 10%        | 1                 | Ⓢ       |
| C507       | ECEA1MU100         | ECEA1HU100           | Electrolytic, 50V, 10μF             | 1                 | Ⓢ       |
| C554       | ECKF1H473ZF        | ECKD1H473ZF          | Ceramic, 50V, 0.047μF, +80%<br>-20% | 1                 | Ⓢ       |
| C571       | ECCD1H151K         | ECCD1H181K           | Ceramic, 50V, 180pF, ± 10%          | 1                 | Ⓢ       |
| C606       | ECEA1AU470         | ECEA1AU331           | Electrolytic, 10V, 330μF            | 1                 | Ⓢ       |
| C713       | Addition           | ECKD1H681KB          | Ceramic, 50V, 680pF, ± 10%          | 1                 | Ⓢ       |
| C811       | Addition           | ECKD1H103ZF          | Ceramic, 50V, 0.01μF, +80%<br>-20%  | 1                 | Ⓢ       |
| C812       | Addition           | ECKD1H103ZF          | Ceramic, 50V, 0.01μF, +80%<br>-20%  | 1                 | Ⓢ       |

## ■ HOW TO CHECK THE PRINTED CIRCUIT BOARD

\* For removal of each part, refer to the Service Manual of SL-P7, (Disassembly instruction on Page 9 ~ 13).

### ● Servo P.C.B.

1. Remove the 4 setscrews of cabinet.
2. Remove the cabinet in the direction of the arrow.
3. Check each part. (Part side)

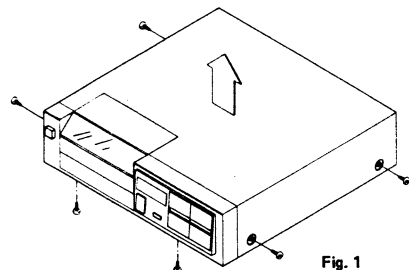


Fig. 1

4. When checking the soldered parts on the servo P.C.B., raise the P.C.B. as in Fig. 2. (Tie the power switch rod to the servo P.C.B. with a string.)

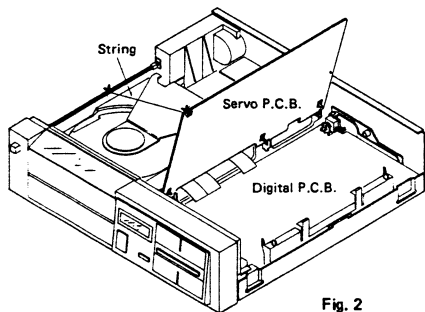


Fig. 2

### ● Fluorescent display tube P.C.B.

1. Remove the cabinet.
2. Remove the switch rod from the power switch.
3. Remove the 4 setscrews of the front panel.

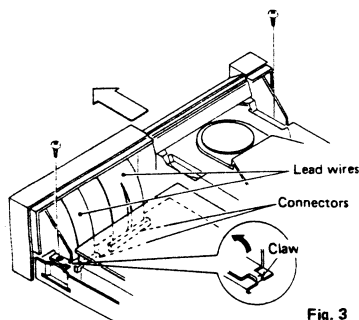


Fig. 3

4. Turn over the front panel and check the P.C.B. as shown in Fig. 4.

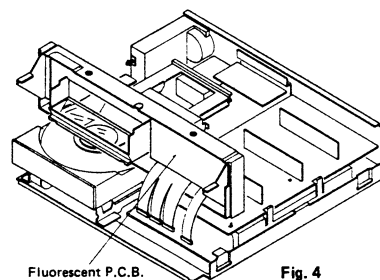


Fig. 4

5. Also remove the 3 setscrews of the P.C.B., and release the 9 claws. (Fig. 5)

6. Shift it as in Fig. 6 and check.

#### Caution:

- (1) Take care not to damage F.L.
- (2) When installing the power switch rod, set it through the lead wire as in Fig. 7.
- (3) When installing cabinet, take care not to squeeze the lead wire.

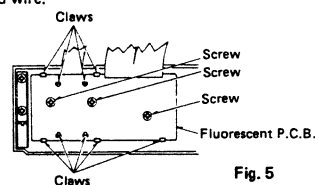


Fig. 5

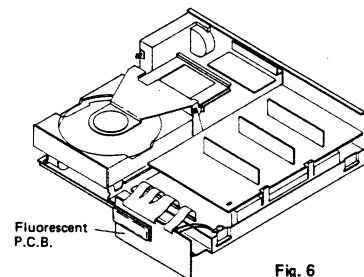


Fig. 6

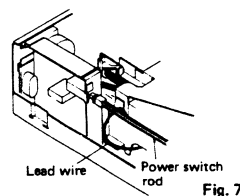


Fig. 7

### ● Digital P.C.B.

1. Remove the cabinet.
2. Remove the servo P.C.B. to shift it as in Fig. 8, and check each part.

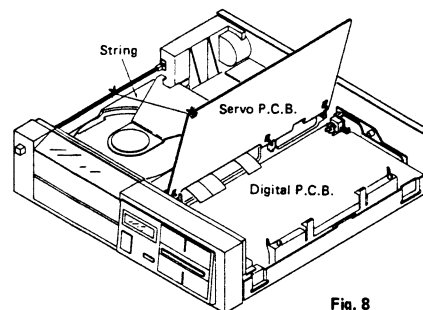


Fig. 8

3. How to check soldered parts:

- (1) Remove the front panel and Fluorescent display tube P.C.B.
- (2) Remove the 3 setscrews of the P.C.B. holder. (Fig. 9)
- (3) Shift it as in Fig. 10 and check.

#### Caution:

When checking as in Fig. 10, connect the ground (earth) terminal of digital P.C.B. to the chassis before turning power supply ON.

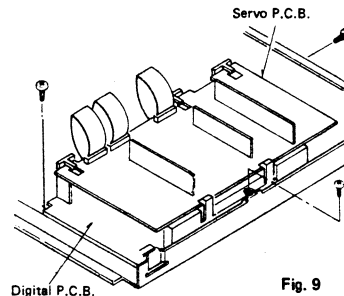


Fig. 9

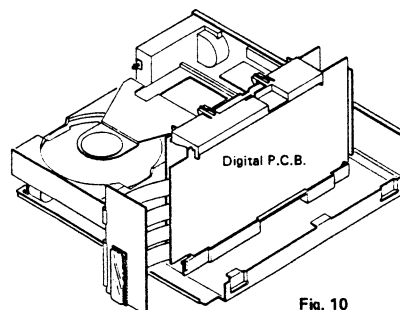
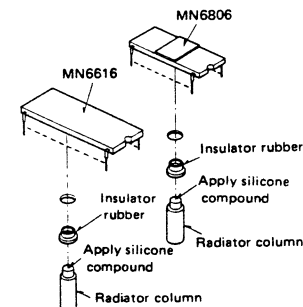


Fig. 10

#### \*Precautions for removal of MN6616 and MN6806

MN6616 and MN6806 attached to the digital P.C.B. are radiated by the radiator column. When fitting LSI, apply silicone compound (or equivalent head diffusing agent) to the radiator column. (Fig. 11)



[The radiator column and insulation (rubber are fixed (glued) on P.C.B.)]

Fig. 11

### ● Head amplifier P.C.B.

\*For the details, refer to the adjustment manual.

1. Remove the cabinet.
2. Remove the front panel, and Fluorescent display tube P.C.B.
3. Pull out the traverse unit connector (attached to the servo P.C.B.). [CN552, CN101 ~ CN106, CN109, earth wire.]
4. Remove the 4 traverse unit setscrews. (Fig. 12)

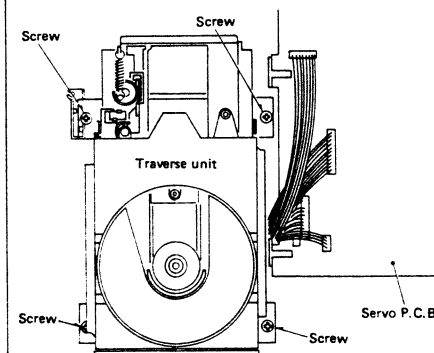


Fig. 12

5. Turn over the traverse unit and remove the 4 nuts (Fig. 13), then the traverse unit ass'y can be removed.
6. Remove the head amplifier P.C.B.
7. Fix the traverse unit ass'y on the jig. (Fig. 14)
8. Check each part.

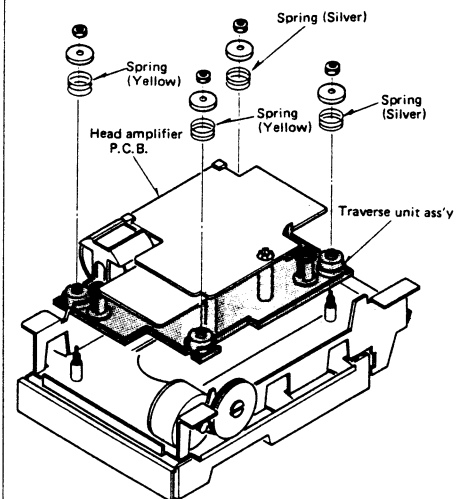


Fig. 13

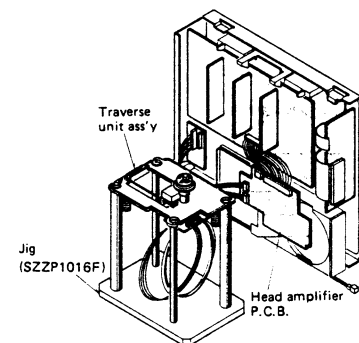


Fig. 14

#### • Spindle motor drive P.C.B.

1. Remove the cabinet.
2. Check each part. (Part side)
3. How to check soldered parts:
  - (1) Remove the switch rod from the power switch.
  - (2) Remove the 2 setscrews of the power transformer. (Fig. 15)
  - (3) Remove the radiator setscrew and take the power transformer block out of the chassis. (Fig. 15)
  - (4) Remove the setscrew and latch. (Fig. 16)
  - (5) Turn over the P.C.B. and check the P.C.B.

#### Caution:

When fitting the P.C.B., be sure to set the insulator sheet.

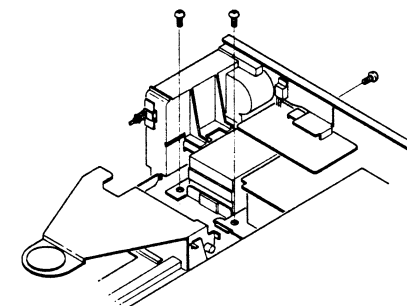


Fig. 15

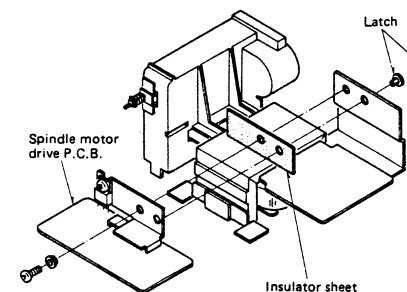
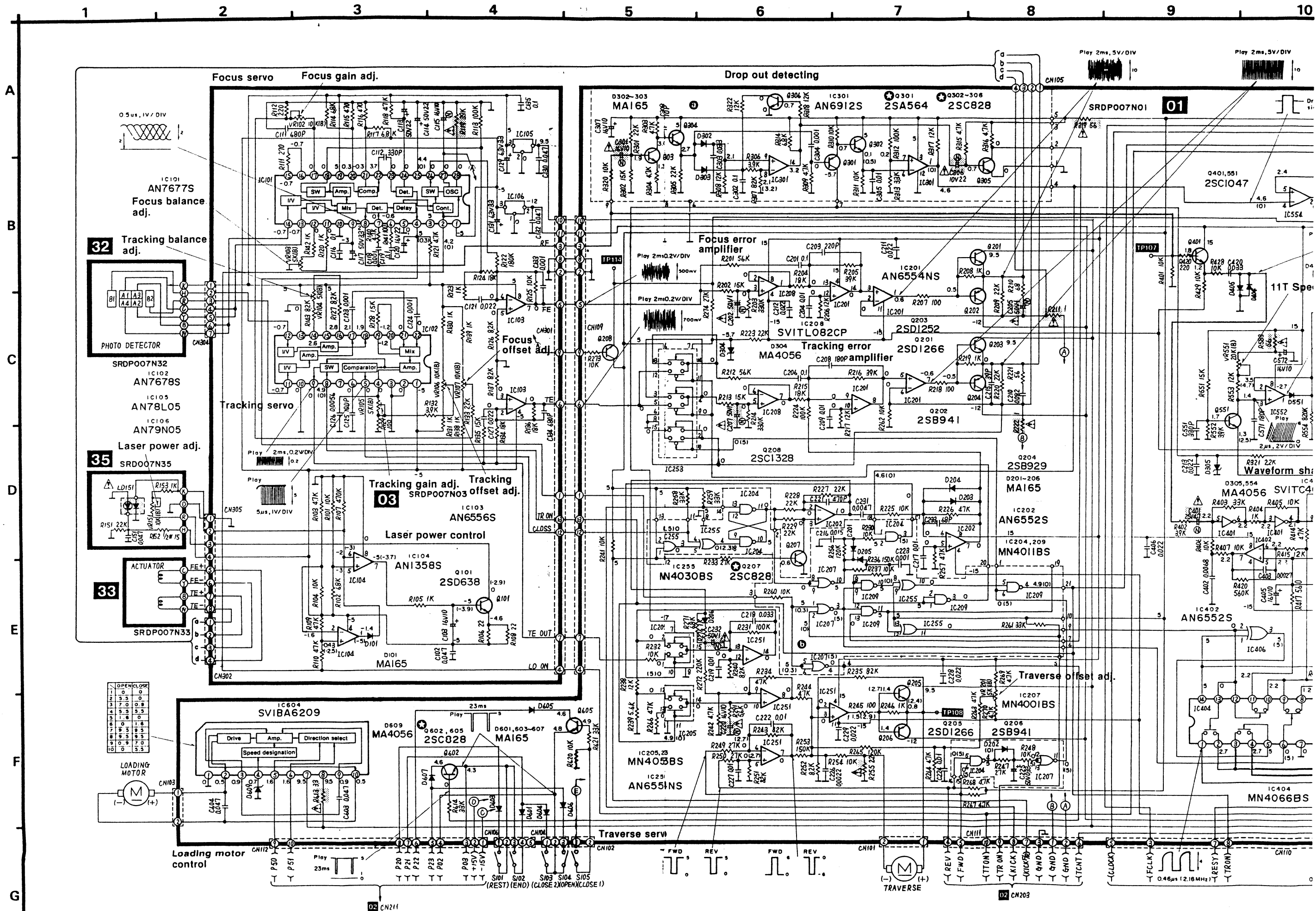
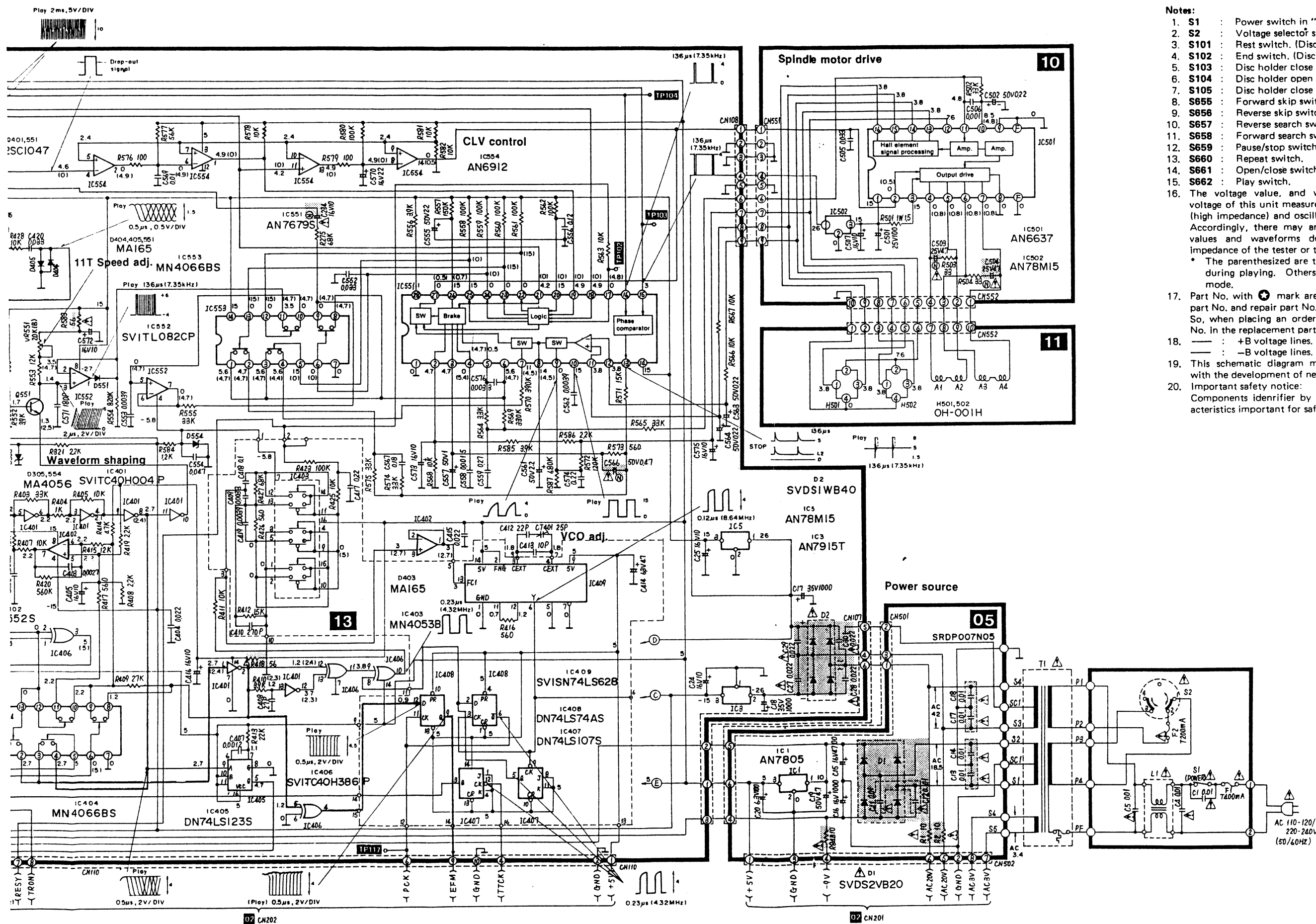


Fig. 16

( This schematic diagram may be modified at any time the development of new technology.)





- Notes:**
- S1** : Power switch in "on" position.
  - S2** : Voltage selector switch.
  - S101** : Rest switch. (Disc Innermost position detection)
  - S102** : End switch. (Disc outside detection)
  - S103** : Disc holder close detection switch.
  - S104** : Disc holder open detection switch.
  - S105** : Disc holder close detection switch.
  - S655** : Forward skip switch.
  - S656** : Reverse skip switch.
  - S657** : Reverse search switch.
  - S658** : Forward search switch.
  - S659** : Pause/stop switch.
  - S660** : Repeat switch.
  - S661** : Open/close switch.
  - S662** : Play switch.
  - The voltage value, and waveforms are the reference voltage of this unit measured by DC electronic voltmeter (high impedance) and oscilloscope on the basis of chassis. Accordingly, there may arise some error in the voltage values and waveforms depending upon the internal impedance of the tester or the measuring unit.
    - The parenthesized are the values of voltage generated during playing. Others are the voltage values in stop mode.
  - Part No. with  mark are not identical between regular part No. and repair part No. supplied.  
So, when placing an order for repair parts, use the part No. in the replacement part list of repair parts.
  - : +B voltage lines.  : Audio signal lines.  
—— : -B voltage lines.
  - This schematic diagram may be modified at any time with the development of new technology.
  - Important safety notice:  
Components identifier by  make have special characteristics important for safety.





01



B

C

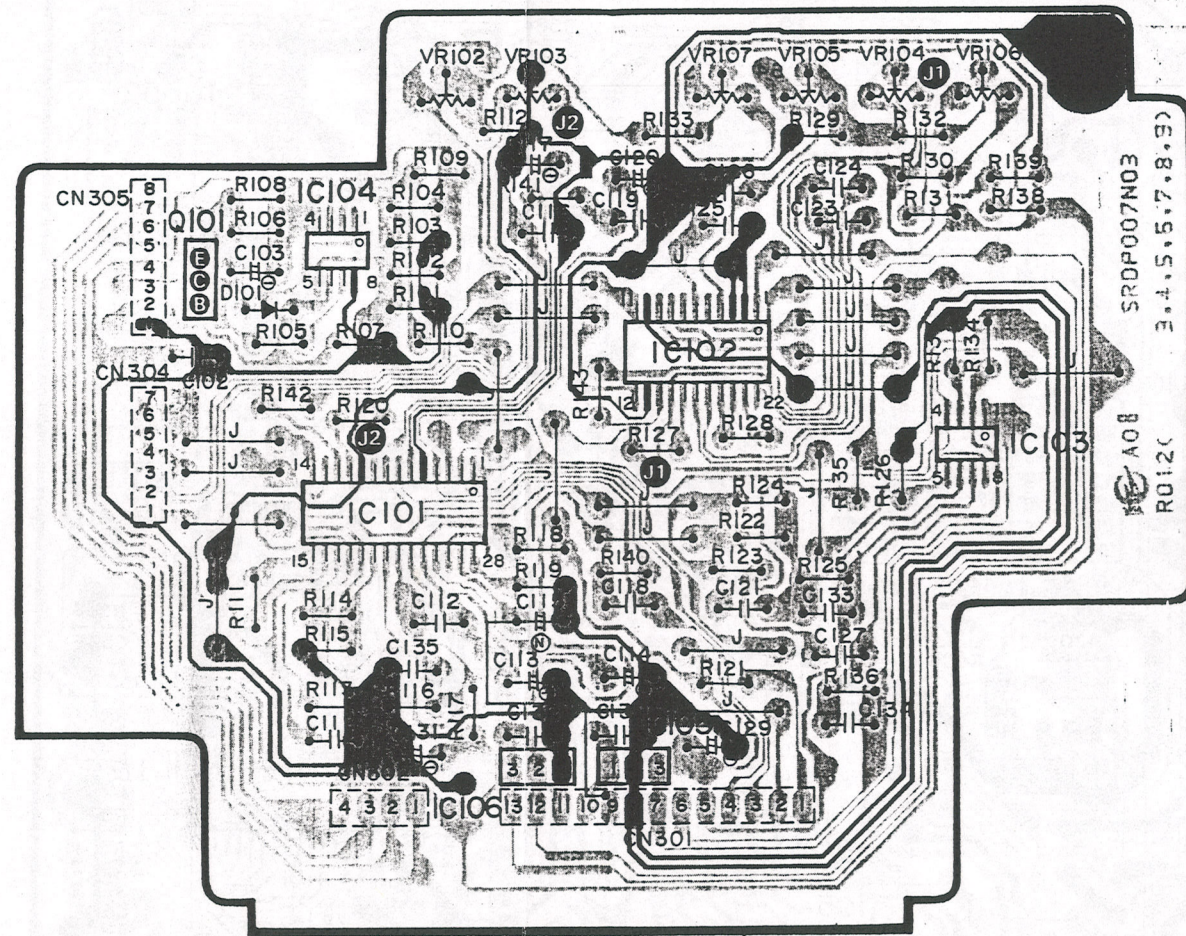
D

E

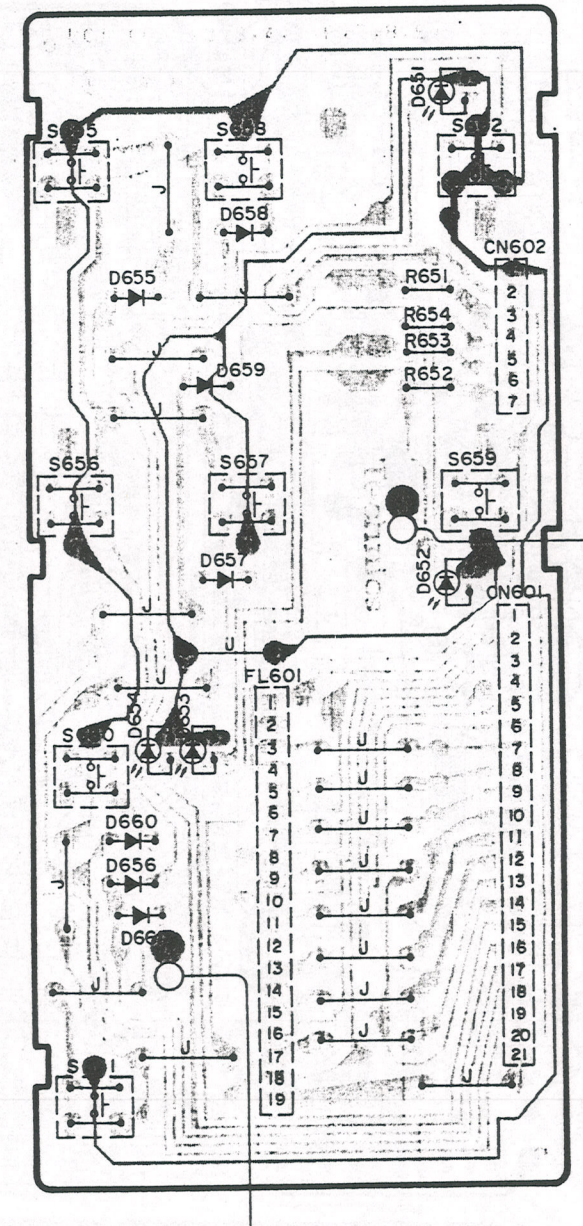


## G

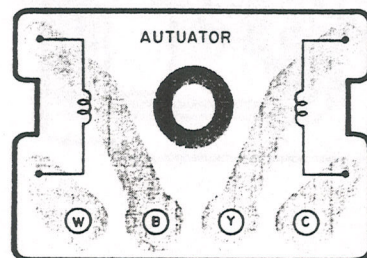
**03** (Head amplifier P.C.B.)



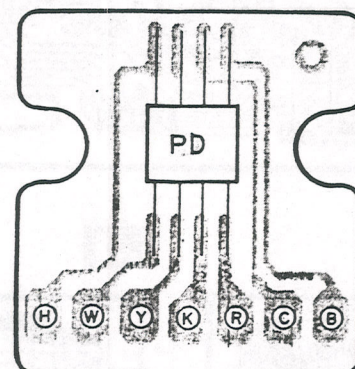
**04** (Display P.C.B.)



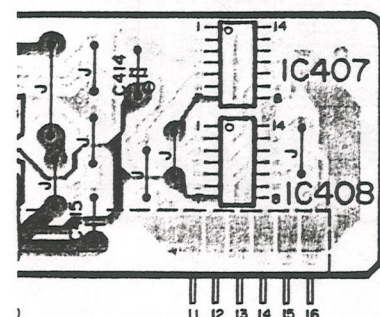
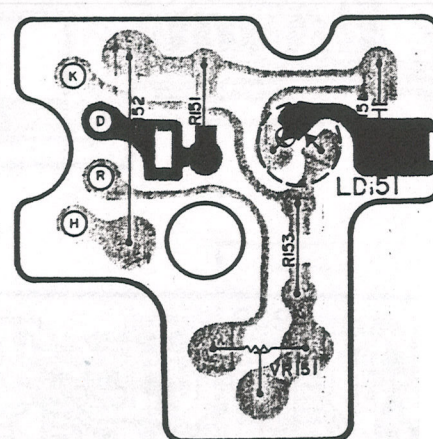
**33** (Actuator coil P.C.B.)

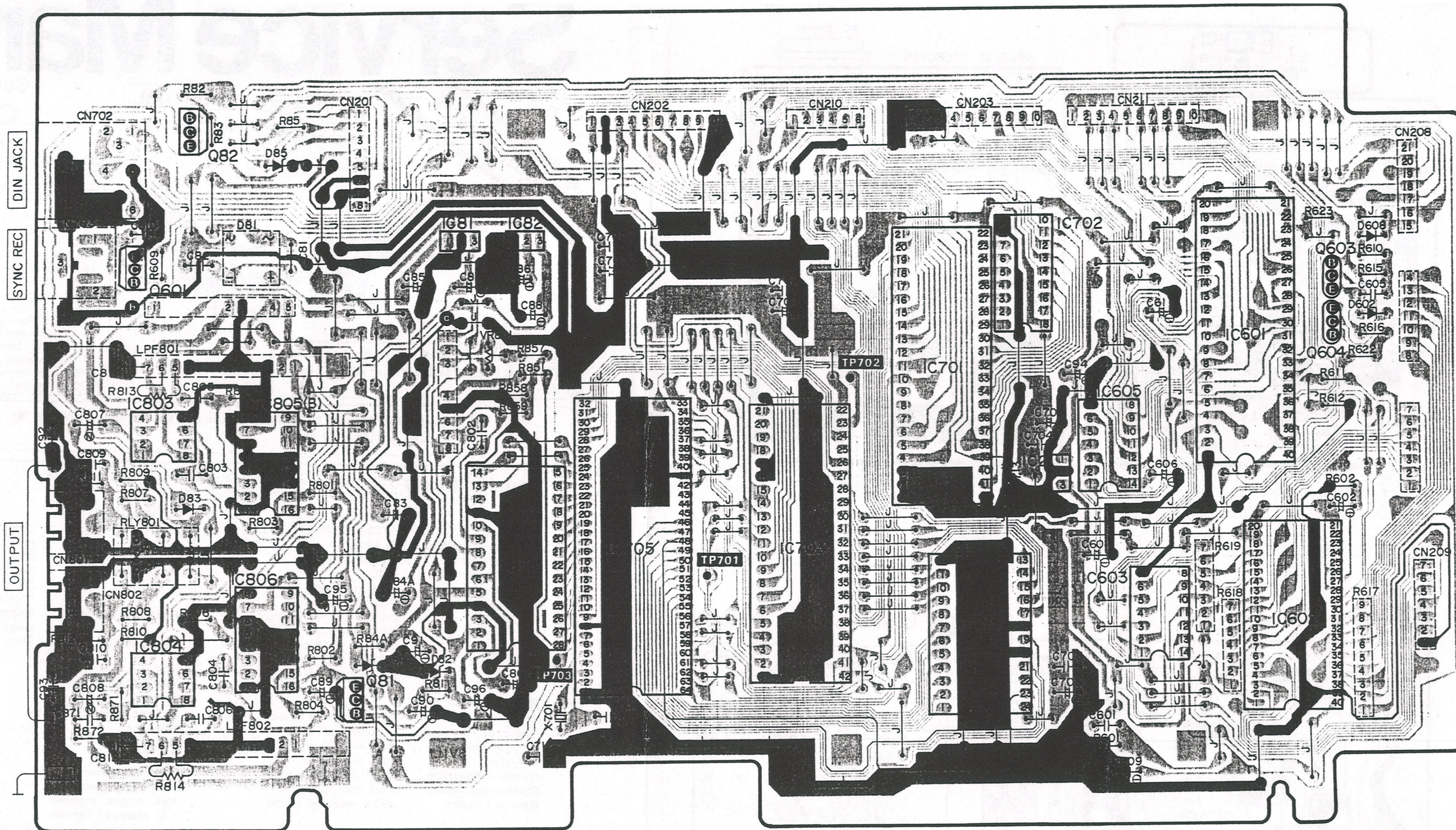


**32** (Photo detector P.C.B.)

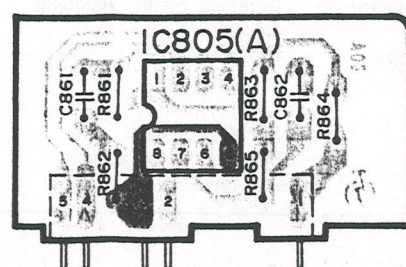


**35** (Laser P.C.B.)

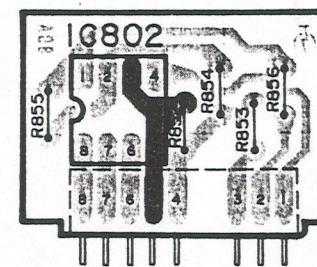




02 b



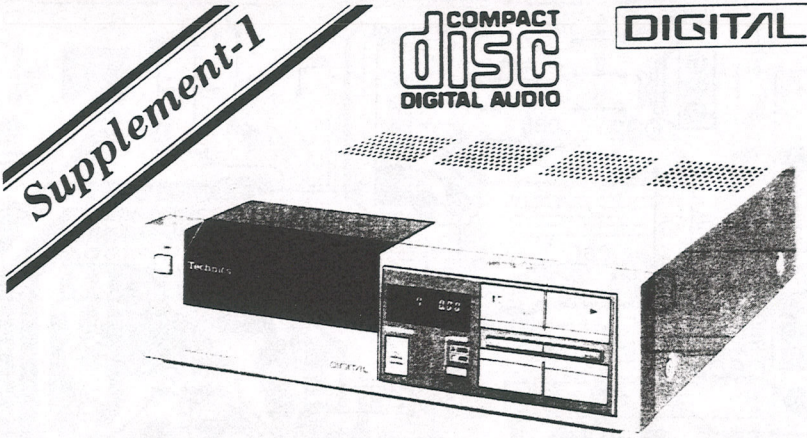
02 c



# Service Manual

Compact Disc Player

## SL-P7/(K)



[E], [EK], [XL], [EG], [EB], [EH], [EF], [Ei], [XA], [PA], [PE], [PC]

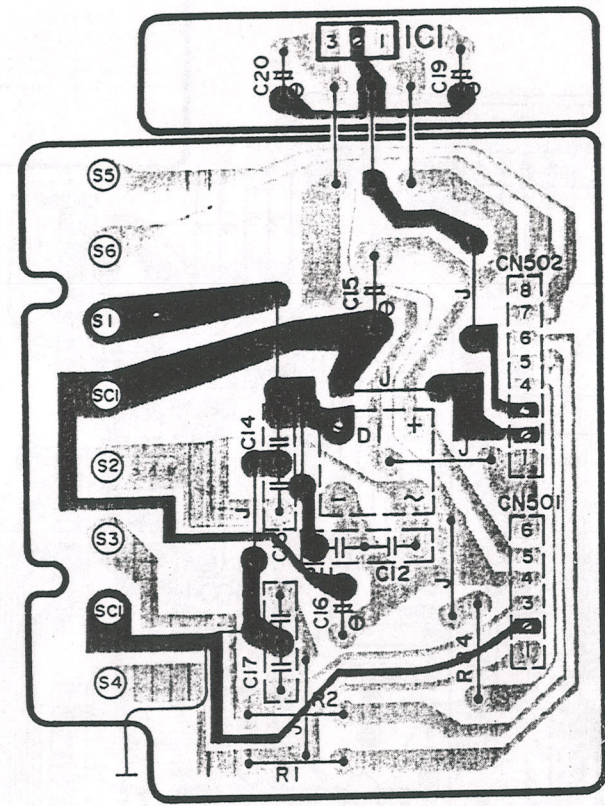
- Areas
- \* [E] is available in Switzerland and Scandinavia.
  - \* [EK] is available in United Kingdom.
  - \* [XL] is available in Australia.
  - \* [EG] is available in F. R. Germany.
  - \* [EB] is available in Belgium.
  - \* [EH] is available in Holland.
  - \* [EF] is available in France.
  - \* [Ei] is available in Italy.
  - \* [XA] is available in Southeast Asia, Oceania, Africa, Middle Near East and Central South America.
  - \* [PA] is available in far East PX.
  - \* [PE] is available in European Military.
  - \* [PC] is available in European Audio Club.

\* The colors of this model include silver and black.  
\* The black type model is provided with (K) in the Service Manual.

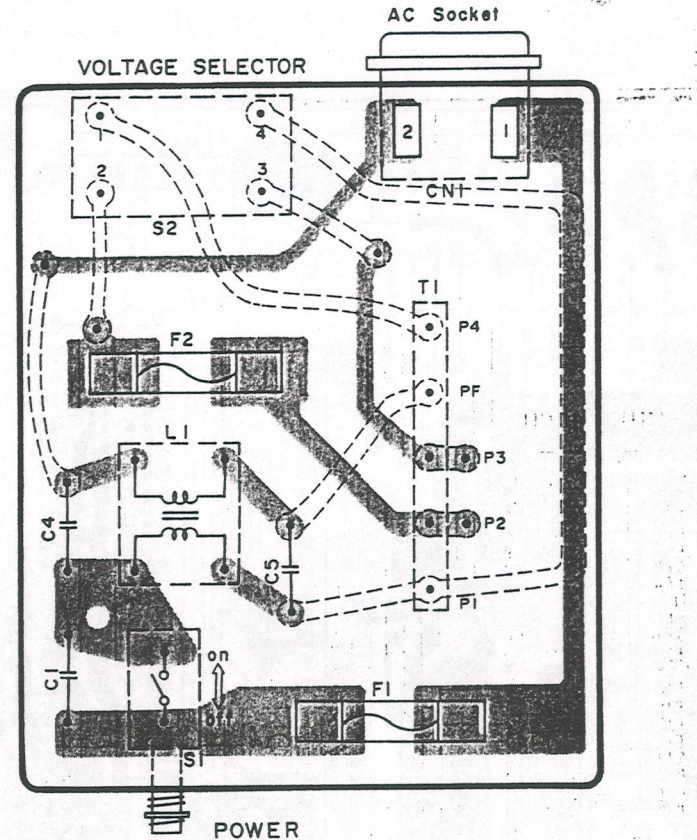
Please use this manual together with the service manual for Model No. SL-P7/(K), Order No. DPD83080021C9.

| Specifications         |                                                                                                                            | Specifications are subject to change without notice for further improvement. Weight and dimensions shown are approximate. |                                                                                                         |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>■ General</b>       |                                                                                                                            | <b>■ Functions</b>                                                                                                        |                                                                                                         |
| Power supply:          | ~110-120/220-240 V, 50/60 Hz                                                                                               | Search modes:                                                                                                             | Manual search                                                                                           |
| Power consumption:     | 30 watts                                                                                                                   |                                                                                                                           | Index search                                                                                            |
| Output voltage:        | 2 volts (at 0 dB)                                                                                                          |                                                                                                                           | Skip search                                                                                             |
| Output impedance:      | 330Ω                                                                                                                       | Display functions:                                                                                                        | No. of tracks, Total playing time, Track being played, Playing time of track being played, Index number |
| Load impedance:        | more than 5 kΩ                                                                                                             | Operation buttons:                                                                                                        | buttons: 7                                                                                              |
| Dimensions (W×D×H):    | 31.5×32.5×8.8 cm (12 <sup>13</sup> / <sub>32</sub> "×12 <sup>13</sup> / <sub>16</sub> "×3 <sup>15</sup> / <sub>32</sub> ") | Disc loading:                                                                                                             | button: 1                                                                                               |
|                        | When disc holder is opened                                                                                                 |                                                                                                                           | Motor driven, lateral loading                                                                           |
|                        | 31.5×45.1×8.8 cm                                                                                                           |                                                                                                                           |                                                                                                         |
|                        | 12 <sup>13</sup> / <sub>32</sub> "×17 <sup>3</sup> / <sub>4</sub> "×3 <sup>5</sup> / <sub>32</sub> "                       |                                                                                                                           |                                                                                                         |
|                        | 4.9 kg (10.8 lbs)                                                                                                          |                                                                                                                           |                                                                                                         |
| <b>■ Audio</b>         |                                                                                                                            | <b>Reference</b>                                                                                                          |                                                                                                         |
| No. of channels:       | 2 (left and right stereo)                                                                                                  | <b>■ Disc specifications</b>                                                                                              |                                                                                                         |
| Frequency response:    | 4-20,000 Hz±0.5 dB                                                                                                         | Diameter:                                                                                                                 | 12 cm                                                                                                   |
| Dynamic range:         | more than 96 dB                                                                                                            | Thickness:                                                                                                                | 0.12 cm                                                                                                 |
| S/N ratio:             | more than 96 dB                                                                                                            | Spindle hole:                                                                                                             | 1.5 cm                                                                                                  |
| Harmonic distortion:   | less than 0.003% (1 kHz, 0 dB)                                                                                             | Min. inside diameter of recorded section:                                                                                 | 5 cm                                                                                                    |
| Channel separation:    | more than 90 dB                                                                                                            | Max. outside diameter of recorded section:                                                                                | 11.6 cm (disc is played from the inside to the outside edge)                                            |
| Wow and flutter:       | below measurable limit                                                                                                     | Direction of rotation:                                                                                                    | Counterclockwise (seen from recorded side of disc)                                                      |
| <b>■ Signal Format</b> |                                                                                                                            | Tracking speed:                                                                                                           | 1.2 to 1.4 meters/second, CLV (Constant Linear Velocity)                                                |
| Sampling frequency:    | 44.1 kHz                                                                                                                   | Rotations per minute:                                                                                                     | About 500 to 200 rpm                                                                                    |
| Correction system:     | Technics Super Decoding Algorithm                                                                                          | Playing time:                                                                                                             | About 60 minutes (up to 75 min. can be contained)                                                       |
| D-A conversion:        | 16-bit linear                                                                                                              | Track pitch:                                                                                                              | 1.6 μm                                                                                                  |
| <b>■ Pickup</b>        |                                                                                                                            | Material:                                                                                                                 | Clear plastic                                                                                           |
| Type:                  | Astigma 3-beam                                                                                                             |                                                                                                                           |                                                                                                         |
| Light source:          | Semiconductor laser                                                                                                        |                                                                                                                           |                                                                                                         |
| Wavelength:            | 800 nm                                                                                                                     |                                                                                                                           |                                                                                                         |

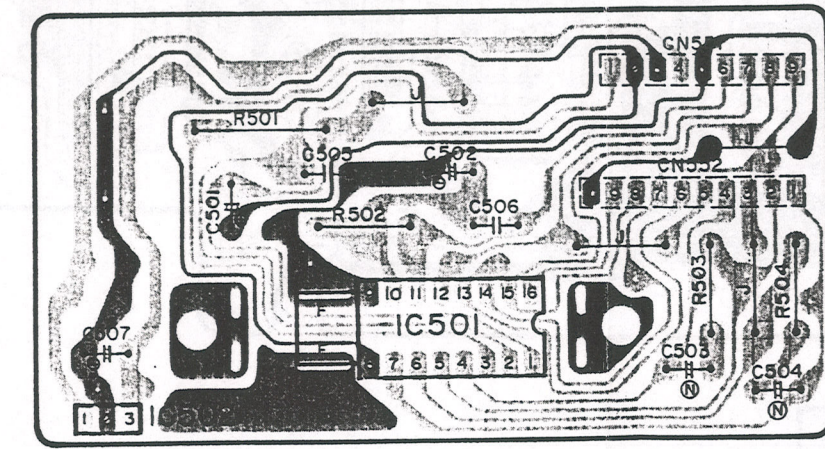
05 (Power supply P.C.B.)



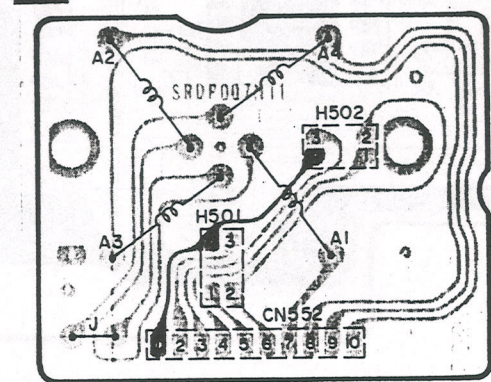
07 (Power fuse P.C.B.) circuit view on top of P.C.B.



10 (Spindle motor drive P.C.B.)

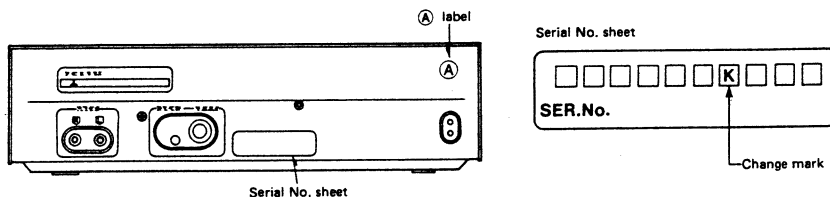


11 (Spindle motor drive coil P.C.B.)



**Note:**

- ★ In order to improve the performance and servicing convenience of SL-P7, discontinuing of P.C.B. ( 12 P.C.B. ; audio circuit, 15 P.C.B. ; CLV circuit), change of P.C.B. pattern diagram ( 01 P.C.B. ; servo circuit, 02 P.C.B. ; digital circuit) and circuit change are conducted during production.
  - ★ The present change is of the sets after the change mark **K** shown in the serial No. sheet provided on the rear chassis of the unit and the carton box.
- Also confirm that the label **A** is provided on AC socket top of rear chassis.



- ★ This supplement refers to after-change P.C.B., schematic diagram, P.C.B. checking method and replacement parts list. The other descriptions are the same as in the service manual (Order No. DPD83080021C9) of SL-P7.
- ★ This supplement is intended to be used together with the service manual (Order No. DPD83080025C1) of SL-P7.

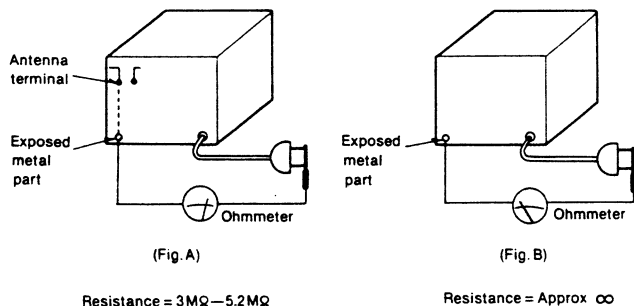
**SAFETY PRECAUTION**

1. Before servicing, unplug the power cord to prevent an electric shock.
2. When replacing parts, use only manufacturer's recommended components for safety.
3. Check the condition of the power cord. Replace if wear or damage is evident.
4. After servicing, be sure to restore the lead dress, insulation barriers, insulation papers, shields, etc.
5. Before returning the serviced equipment to the customer, be sure to make the following insulation resistance test to prevent the customer from being exposed to a shock hazard.

**INSULATION RESISTANCE TEST**

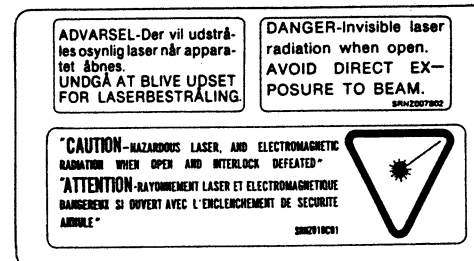
1. Unplug the power cord and short the two prongs of the plug with a jumper wire.
2. Turn on the power switch.
3. Measure the resistance value with ohmmeter between the jumpered AC plug and each exposed metal cabinet part, such as screwheads antenna, control shafts, handle brackets, etc. Equipment with antenna terminals should read between  $3M\Omega$  and  $5.2M\Omega$  to all exposed parts. (Fig. A) Equipment without antenna terminals should read approximately infinity to all exposed parts. (Fig. B)

**Note:** Some exposed parts may be isolated from the chassis by design. These will read infinity.



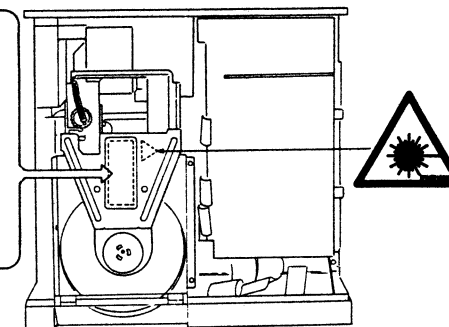
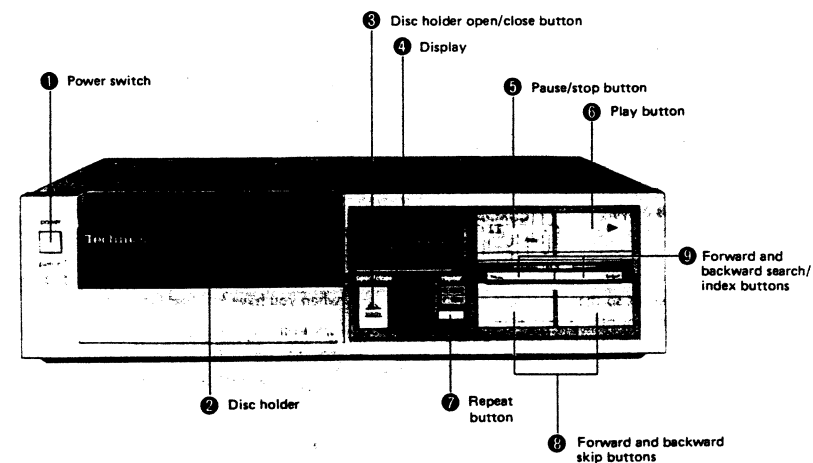
4. If the measurement is outside the specified limits, there is a possibility of a shock hazard. The equipment should be repaired and rechecked before it is returned to the customer.

Caution : This product utilizes a laser diode.

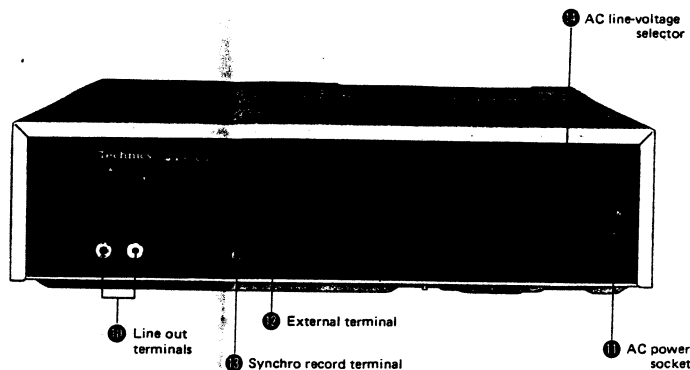
**Use of caution labels**

|            | U.S.A. | Canada | Europe | Others |
|------------|--------|--------|--------|--------|
| SRNZ010S01 | X      | X      | ○      | ○      |
| SRNZ007S02 | ○      | X      | ○      | ○      |
| SRNZ010C01 | X      | ○      | X      | X      |
| SRNZ010S02 | X      | X      | ○      | ○      |

Note: ○ Mark Label is used. X Mark Label is not used.

**LOCATION OF CONTROLS****Front panel view**

## • Rear panel view



## Functions

### 1 Power switch

- Press once to turn power on and again to turn power off.
- When power is turned on, the display ④ will read ( 0 ) for the track number and ( 000 ) for the time.
- If no disc is in the holder, the play button ⑥ ( ▶ ) will only light up for about five seconds.

### 2 Disc holder

- The disc is inserted in this holder.
- Press the disc holder open/close button ③ to open the holder and press the open/close button again to close the holder.
- The holder is also closed automatically when the play button ⑥ is pressed even when the open/close button is not pressed a second time. (The holder can also be closed by hand.)

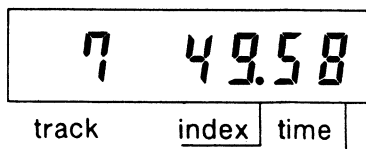
### 3 Disc holder open/close button

- Press this button to insert or remove a disc. When this button is pressed, the disc holder ② opens.
- Press this button again to close the holder.

### 4 Display

The display consists of the following items:

- Track number (up to 99)
- Index number—Displays the minutes portion of the total playing time or the index number. Note that the index number is displayed only when the unit is in the index play mode.
- Time—Displays the seconds portion of the total playing time except when the index number is being displayed.



- The unit switches to the standby mode when a disc is inserted in the holder. In the standby mode, the pickup is located at the beginning of the first track and the display is showing the total number of tracks on the disc and the total playing time.

- The above example shows the display for a disc having seven tracks and a total playing time of 49 minutes 58 seconds.

- The number of tracks and total playing time is also displayed when the play button ⑥ is pressed after inserting a disc (without pressing the open/close button ③ to close the disc holder ②). In this case, disc play will automatically begin from the first track.

### 5 Pause/stop button

- This button activates both the pause and stop modes.
- When this button is pressed while a disc is being played, the II indicator lights to show that the pause mode has been activated. Press the play button ⑥ to cancel the pause mode and resume disc play.
- If the pause/stop button ⑤ is held down for more than about two seconds, the stop mode is activated (the II indicator goes out) and the pickup returns to the beginning of the first track on the disc.
- The disc continues to rotate for about 20 seconds after the stop mode is activated so that the player can quickly switch to the play or skip function if one of those buttons is pressed. If these buttons are not pressed during the 20 seconds period, the disc will stop rotating.

### 6 Play button

- Press this button to begin disc play. The play indicator ( ▶ ) lights up to show when this button has been pressed.
- When this button is pressed during disc play, the pickup returns to the starting position and disc play starts again from the beginning.
- When this button is pressed while the pause mode is activated, pause is cancelled and the player returns to the play mode.

### 7 Repeat button

- Press this button to activate the repeat play mode.
- When the [REPEAT] indicator lights, only the current track will be played repeatedly.
- When the [ALL] indicator lights, the entire side of the disc will be played repeatedly.
- When neither indicator is on, the repeat mode has been cancelled.

### 8 Forward and backward skip buttons

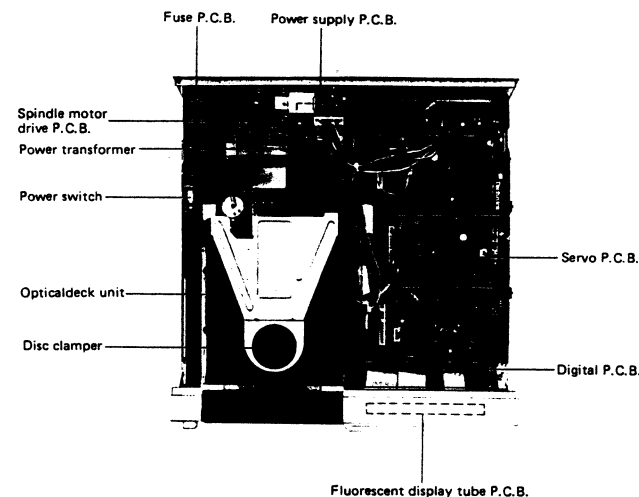
- Use these buttons to move the pickup forward and backward to the beginning of a specific track. The buttons are used in the following manner.

1. For direct access play to locate the beginning of a specific track when the player is in the stop or pause mode.
  2. For skip play to skip one or more track when the player is in the play or pause mode.
- Press the [SKIP FWD] button to move the pickup forward.
  - Press the [SKIP BWD] button to move the pickup backward.
  - When either of these buttons is held down, tracks are skipped continuously. (Quick skip)

### 9 Forward and backward search/index buttons

- Use these buttons to move the pickup forward and backward. The buttons are used in the following manner.
1. For manual search play by moving the pickup to a specific location when the player is in play or pause mode.
  - Press the [SEARCH FWD] button to move the pickup forward.
  - Press the [SEARCH BWD] button to move the pickup backward.
  - When either of these buttons is pressed, the pickup will move slowly at first and then rapidly if the button is held down for more than about three seconds.
  2. For index play when the player is in the stop mode.

### • Top view



### 10 Line out terminals (LINE OUT)

- These are the audio output terminals.
- Connect the line out terminals to the AUX or TAPE PLAYBACK terminals on your amplifier or receiver.

### 11 AC power socket (AC IN)

Connect this socket to a wall socket using the power cord.

### 12 External terminal

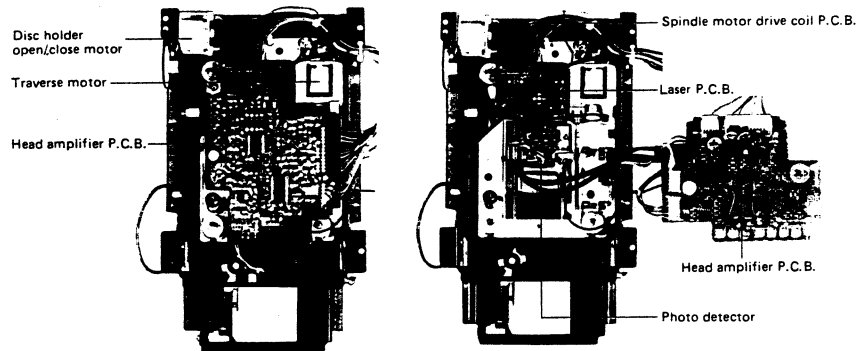
### 13 Synchro record terminal

The following functions are possible when this terminal is used:

1. Recording can be synchronized with the beginning and end of disc play and pause operation. Note that this is only possible when using Technics cassette decks which are equipped with a SYNCHRO REC terminal. (Synchro recording function)
2. The amplifier input source can be automatically switched to aux/CD/video when the play button ⑥ is pressed. It is also possible to automatically start disc play when the amplifier input source selector is switched to aux/CD/video. (Direct operation function)

### 14 AC line-voltage selector

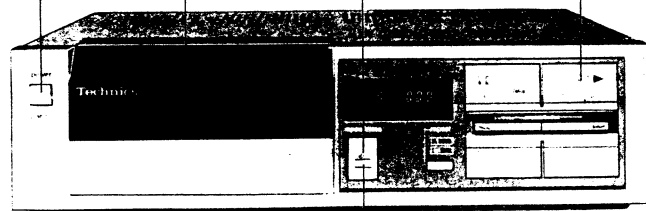
### • Optical deck unit (Bottom view)



## ■ PLAYING A DISC

### 1. Automatic Play: (To play a disc from the beginning)

1. Press the power switch to the on position. (I →)
3. Insert a disc.
4. Press the disc holder open/close button again.
5. Press the play button.



2. Press the disc holder open/close button.

#### • To briefly interrupt disc play

- Press the pause/stop button to activate the pause mode (II indicator lights). Be careful not to hold the pause/stop button down more than about two seconds since this will switch the player to the stop mode.
- Press the play button to resume disc play.

#### • To stop disc play

- Hold the pause/stop button down for about two seconds until the II indicator lights off.
- Turn the power switch off if the player is not going to be used for a while.

## 2. Direct Access Play

Direct access play can be performed when the player is in the stop or pause mode.

- For example, when the third track is specified.

| Procedure                                  | Display                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| skip<br>Press three times.                 | The pickup moves to the beginning of track 3 and the track indicator displays the number 3.<br> |
| skip<br>Press to move the pickup backward. |                                                                                                 |
| play<br>Press to start disc play.          | 3 0.01<br>Track No.      Playing time mode                                                      |

## 3. Index Play

Index play can be performed when the player is in the stop mode.

- Use index play to play a specific index of the disc as listed in index of a Compact Disc's instruction sheet.  
Example: To play the second index of the third track which contains a total of five indexes.

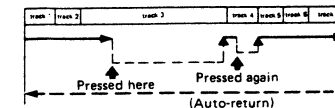
| Procedure                                              | Display                                   |
|--------------------------------------------------------|-------------------------------------------|
| skip<br>While watching the display, enter track no. 3. | 3<br>Track No.                            |
|                                                        | Specify index no. 2.<br>3 02<br>Index No. |
| play<br>Press to start disc play.                      | 3 4.21<br>Playing time mode               |

## 4. Skip Play

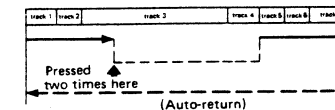
Skip play can be performed when the player is in the play or pause mode.

### • Forward skip play

Example 1: When the forward skip button is pressed while listening to the third track to advance to the beginning of the fourth track and the again to advance to the beginning of the fifth track.

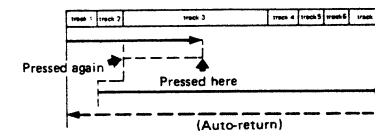


Example 2: To skip directly from the third track to the beginning of the fifth track, press the forward skip button two times in a row. (quick skip)

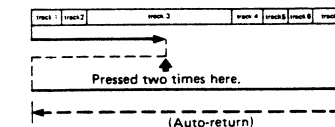


### • Backward skip play

Example 1: When the backward skip button is pressed while listening to the third track to return to the beginning of the track and then again to return to the beginning of the second track.



Example 2: To skip directly from the third track to the beginning of the second track, press the backward skip button two times in a row. (quick skip)



#### Notes:

- In both the forward and backward skip modes, the number of tracks skipped is the same as the number of times either skip button is pressed. Remember that for backward skip, the track currently being played is counted as the first track.
- Press the skip buttons while watching the track number display.
- If skip play is performed while the player is in the pause mode, press the play button to resume disc play after the pickup has moved to the desired point.

## 5. Manual

Manual player is

- To move the pickup
- Press the move the pickup



- The pickup moves more than about two seconds.
- When the play button is pressed, the play button will not respond.
- During the desired track.

#### Note:

- When the pickup is moved to the desired track, the pickup will not respond.

## ■ HAN

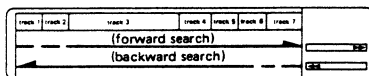
- Handling the compact disc
- Only compact disc can be played.

- Hold compact disc surface is

## 5. Manual Search Play

Manual search play can be performed when the player is in the play or pause mode.

- **To move the pickup to a specific point**
- Press the forward or reverse search/index button to move the pickup to the desired point.



- The pickup will move slowly at first and then switch to rapid movement if the button is held down for more than about three seconds.
- When manual search is performed while the player is in the play mode, the desired point can be located by watching the display. (at a level of about -12 dB of the present volume setting).
- During the pause mode, watch the display to locate the desired point on the disc.

### Note:

- When the player is in the single track repeat mode, the pickup can only be moved to locations within that track.

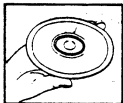
## ■ HANDLING COMPACT DISCS

### ● Handling precautions

Only compact discs identified with this mark can be used with this player.



- Hold compact discs by the edges so the surface is not soiled with fingerprints.



- Be careful not to scratch the surface with fingernails or other sharp objects, particularly when inserting and removing discs in their cases.
- Do not bend the disc.
- Do not use record cleaning sprays or static electricity prevention liquids.
- Do not wipe with benzene, thinner or any other solvent. If the surface is soiled, wipe gently with a soft damp (water only) cloth.

## 6. Repeat Play

### 1. To repeatedly play an entire side:

Press the repeat button until the [REPEAT] indicator lights.

### 2. To repeatedly play one track:

Press the repeat button until the [TRACK] indicator lights.

- The repeat play mode can be activated while a disc is being played.

- Follow directions in the direct access section, if you want to specify the track to be played repeatedly.

### 3. To cancel the repeat mode:

To stop repeat play, press the repeat button until all repeat indicators are out or press the pause/stop button to stop disc play.

### Note:

Repeat play can only be done for an entire disc side or an entire track. It can not be set by index numbers. Therefore, on discs having only one track, the entire side and single track repeat modes will both play the entire side repeatedly.

## 7. Timer Play

1. Turn on the power switches of all components in the timer play chain.

**Note:** The timer of the audio timer should be off at this time.

2. Set the amplifier input selector to the position for this player.
3. Insert a disc in the holder.
4. Confirm the present time and set the audio timer to the desired start and stop times.
5. Turn the audio timer on. The audio timer will now turn power off to all components in the timer play chain.
6. The disc will now be played from track one at the preset time.

- If the disc is brought from a cold environment into a warm room, dew may form on the disc. Wipe this off with a soft, dry cloth before using the disc.
- Do not write on the label with a ball-point pen, hard pencil or other hard writing utensil.
- Always remove the disc from the disc compartment when you have finished listening to it.

### ● Storage precautions

- Be sure to store discs in their cases to protect them from dust, scratches and warping.
- Do not place or store discs in the following places:
  - 1) Locations exposed to direct sunlight.
  - 2) Locations with high humidity or a lot of dust.
  - 3) Locations directly exposed to a heat outlet or heating appliance.
  - 4) In the glove compartment or rear ledge of an automobile.

## ■ DISASSEMBLY INSTRUCTIONS

### ● How to remove the cabinet

1. Remove the 4 setscrews [Fig. 1 ; ① ~ ④] to remove the cabinet in the direction of the arrow.

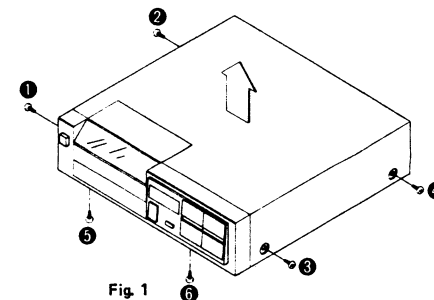


Fig. 1

### ● How to remove the front panel

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Remove the 2 setscrews [Fig. 1 : ⑤, ⑥] of the front panel.
3. Remove the 2 setscrews [Fig. 2 : ⑦, ⑧] of the front panel fitting, and release the 2 claws.
4. Pull out the lead wires from the connectors of the digital P.C.B. and remove the front panel in the direction of the arrow.

### Note:

When installing the front panel, remove the power switch rod and install the front panel from the disc holder side. (Take care not to scratch the disc holder.)

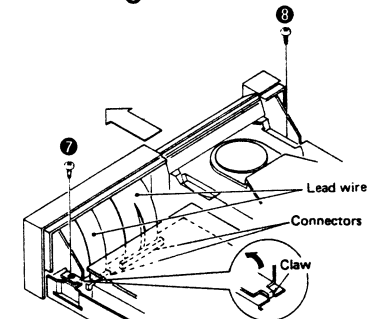


Fig. 2

### ● How to remove the fluorescent display tube P.C.B.

1. Remove the front panel. [Refer to "How to remove the front panel".]
  2. Remove the 3 setscrews [Fig. 3 : ⑨ ~ ⑪] of the fluorescent display tube P.C.B. and release the 9 claws, then the fluorescent display tube P.C.B. can be removed.
- \*Open/close, repeat button and search/index button can be removed by removing the fluorescent display tube P.C.B.

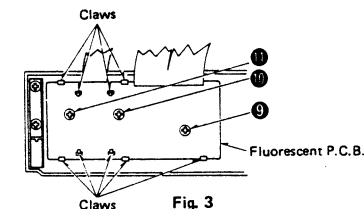


Fig. 3

### ● How to remove the servo P.C.B.

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Release the 2 claws [Fig. 4] in the direction of the arrow and then lift the servo P.C.B.
3. Push the 3 claws [one by one] in the direction of the arrow (A) in Fig. 5, and remove the servo P.C.B. by pulling it in the direction of the arrow (B).
4. To remove the audio P.C.B., remove the setscrew ⑫ [Fig. 4].

\*When checking the circuit, do the job as in paragraph 2.

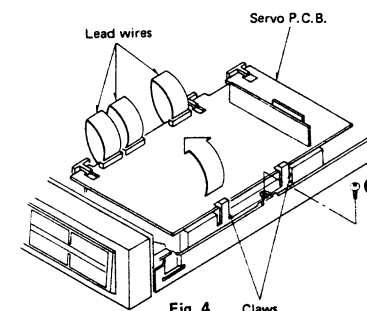


Fig. 4

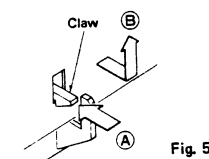


Fig. 5

## DISASSEMBLY INSTRUCTIONS

### How to remove the cabinet

1. Remove the 4 setscrews [Fig. 1 : ① ~ ④] to remove the cabinet in the direction of the arrow.

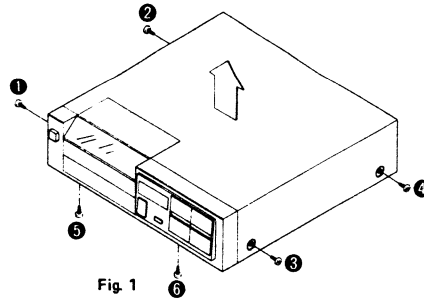


Fig. 1

### How to remove the front panel

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Remove the 2 setscrews [Fig. 1 : ⑤, ⑥] of the front panel.
3. Remove the 2 setscrews [Fig. 2 : ⑦, ⑧] of the front panel fitting, and release the 2 claws.
4. Pull out the lead wires from the connectors of the digital P.C.B. and remove the front panel in the direction of the arrow.

#### Note:

When installing the front panel, remove the power switch rod and install the front panel from the disc holder side. (Take care not to scratch the disc holder.)

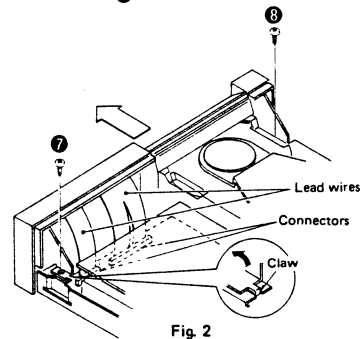


Fig. 2

### How to remove the fluorescent display tube P.C.B.

1. Remove the front panel. [Refer to "How to remove the front panel".]
2. Remove the 3 setscrews [Fig. 3 : ⑨ ~ ⑪] of the fluorescent display tube P.C.B. and release the 9 claws, then the fluorescent display tube P.C.B. can be removed.

\*Open/close, repeat button and search/index button can be removed by removing the fluorescent display tube P.C.B.

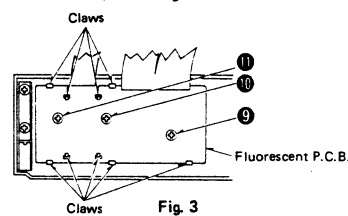


Fig. 3

### How to remove the servo P.C.B.

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Release the 2 claws [Fig. 4] in the direction of the arrow and then lift the servo P.C.B.
3. Push the 3 claws [one by one] in the direction of the arrow (A) in Fig. 5, and remove the servo P.C.B. by pulling it in the direction of the arrow (B).
4. To remove the audio P.C.B., remove the setscrew ⑫ [Fig. 4].

\*When checking the circuit, do the job as in paragraph 2.

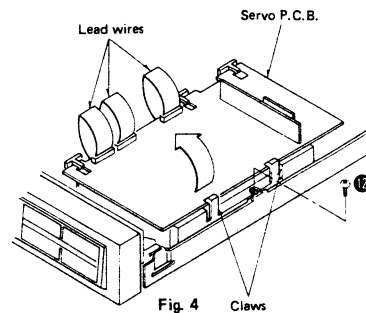


Fig. 4

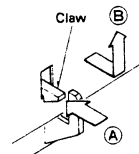


Fig. 5

### How to remove the digital P.C.B.

1. Remove the servo P.C.B. [Refer to "How to remove the servo P.C.B."]
2. Remove the 2 setscrews [Fig. 6 : ⑬, ⑭] of P.C.B. holder.
3. Remove the setscrew [Fig. 6 : ⑮] of the output jack holder.
4. Remove the P.C.B. (with holder) in the direction of the arrow. [Fig. 6]
5. To remove the digital P.C.B., release the 9 claws of each holder.

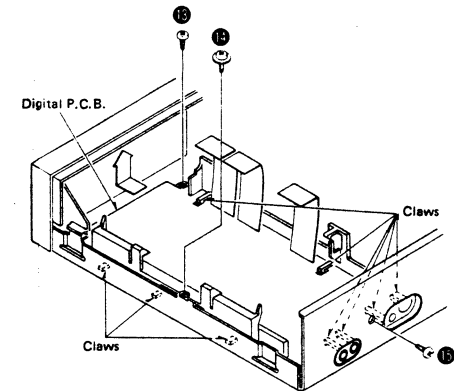


Fig. 6

### How to remove the power transformer

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Pull out the disc holder in the direction of the arrow [Fig. 7].
3. Remove the setscrew [Fig. 7 : ⑯] of the spindle motor drive P.C.B.
4. Remove the 2 setscrews [Fig. 7 : ⑰ ~ ⑱] of the power transformer.
5. Remove the radiator setscrew [Fig. 7 : ⑲] and then remove the transformer block.
6. Remove the 2 setscrews [Fig. 7 : ⑳, ㉑] of power supply P.C.B.
7. Unsolder the 8 portions of power supply P.C.B. to remove the P.C.B.
8. Remove the switch rod from the power switch.
9. Remove the 2 setscrews [Fig. 7 : ㉒, ㉓] of fuse P.C.B. holder.
10. Unsolder the 5 portions of fuse P.C.B. and remove the fuse P.C.B. from the power transformer along with the holder.

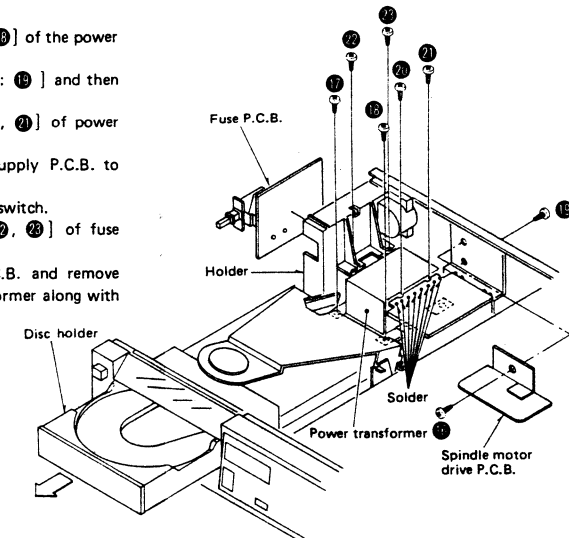


Fig. 7

### How to remove disc clamper

1. Remove the cabinet. [Refer to "How to remove the cabinet".]
2. Pull out the disc holder.
3. Remove the disc clamper setscrew [Fig. 8 : ㉔] and remove the spring.
4. Remove the disc clamper in the direction of the arrow [Fig. 8].

#### Notes:

1. Be sure to pull out the disc holder and remove the disc clamper.
2. Do not remove the stopper. If it is removed, re-adjustment will be needed.

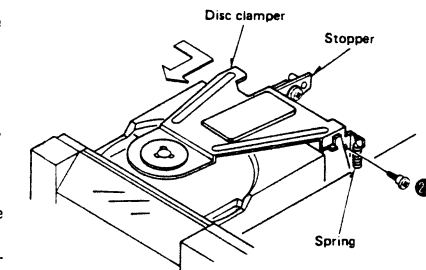


Fig. 8

### • How to remove the disc holder

1. Remove the front panel. [Refer to "How to remove the front panel".]
2. Remove the disc clasper. [Refer to "How to remove the disc clasper".]
3. Remove the disc holder setscrews [Fig. 9 ; ②, ⑦].
4. Disconnect the lead wire from the lead wire holder. [Take care not to break the lead wire.]
5. Push the disc holder in the direction of arrow (A) and release the 2 claws from the bottom, then remove the disc holder by lifting it in the direction of arrow (B).

\*Fit the disc holder with the drive rack pressed in the direction of arrow (C).

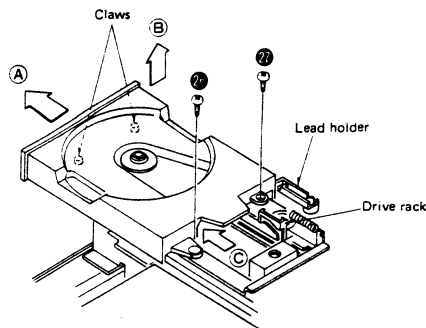


Fig. 9

### • How to remove and set the disc tray

1. Remove the disc holder. [Refer to "How to remove the disc holder".]
2. Release the 2 claws from the bottom of disc holder and then remove the disc tray. [In this case, be careful not to lose the 3 ball bearings.]
3. When setting the disc tray, accurately fit the lever onto the boss as illustrated.

\*It can be easily set by pushing the lever in the direction of the arrow.

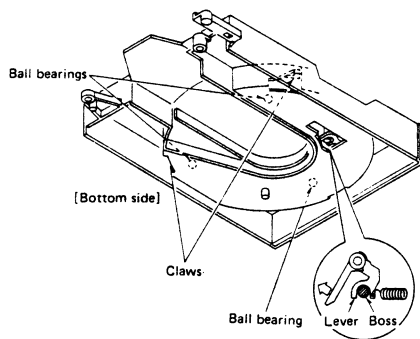


Fig. 10

### • How to remove the loading guide

1. Remove the disc holder. [Refer to "How to remove the disc holder".]
2. Release the 2 claws of close switch cover to release the switch. (Fig. 11)
3. Remove the 6 setscrews [Fig. 11 ; ②⑤ ~ ③①] on the right and left sides of loading guide holder. Then the loading guide can be removed.

\*When fitting the close switch and return gear again, push the drive rack in the direction of the arrow and set them so that the return gear spring is set on the left side. [See Fig. 12.] (The — marked gear of return gear is engaged between the 2nd and 3rd from top of the drive rack gears.)

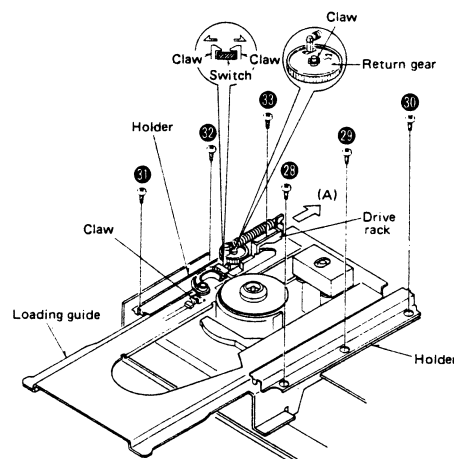


Fig. 11

### • How to remove the drive rack

1. Remove the loading guide. [Refer to "How to remove the loading guide".]
2. Release the return gear claw to remove the return gear. [Fig. 11]
3. Push the drive rack in the direction of arrow (A), and release the drive rack claw, then the drive rack can be removed downward. [Fig. 11]

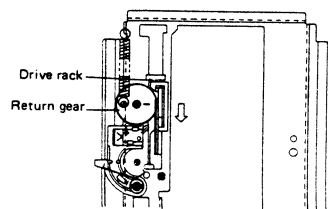


Fig. 12

### • How to remove the traverse unit [Optical deck unit]

1. Remove the front panel. [Refer to "How to remove the front panel".]
2. Remove the 4 setscrews [Fig. 13 ; ④ ~ ⑦] of traverse unit. Then the traverse unit can be removed.
3. Pull out the traverse unit connector (attached to the servo P.C.B.). [CN552, CN101~CN106, CN109, earth wire.]
4. Turn over the traverse unit and remove the 4 nuts [Fig. 14 ; ① ~ ④]. Then the travers unit ass'y.

Replacement parts are included in the traverse unit ass'y (optical deck unit) supplied.  
So, do not remove the limit switch, motor, turntable and pickup.

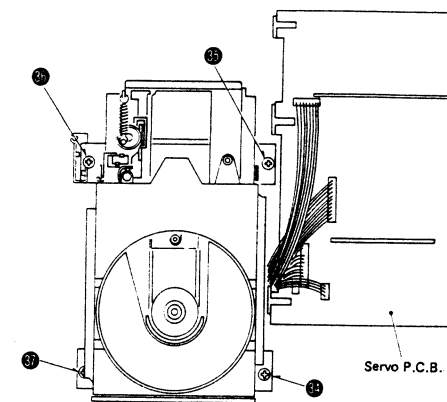


Fig. 13

### • How to remove the head amplifier P.C.B. and spindle motor drive coil P.C.B.

1. Remove the 4 setscrews [Fig. 13 ; ④ ~ ⑦] of traverse unit.
2. Turn over the traverse unit and remove the head amplifier setscrew [Fig. 14 : ⑤].
3. Disconnect the lead wire from the lead wire holder.
4. Release the claw from the head amplifier P.C.B. Then the head amplifier P.C.B. can be removed.
5. Remove the 2 setscrews [Fig. 14 ; ③, ④] of spindle motor P.C.B. Then the spindle motor drive coil P.C.B. can be removed.

### • How to remove the limit switch [Rest and end detecting switch]

1. Remove the head amplifier P.C.B. [Refer to "How to remove the head amplifier P.C.B. and spindle motor P.C.B.".]
  2. Remove the 2 setscrews [Fig. 14 : ③, ④] of limit switch, and unsolder the limit switch.
- \*After fitting the limit switch [rest : disc's innermost position detection] again, it is necessary to re-adjust the position.

### • How to remove the loading motor

1. Remove the head amplifier P.C.B. [Refer to "How to remove the head amplifier P.C.B. and spindle motor P.C.B.".]
  2. Remove the belt from the loading motor pulley. [Fig. 14]
  3. Remove the 2 setscrews [Fig. 14 : ③, ④] of loading motor, and unsolder the motor leads.
- \*Connect the brown lead to the (+) terminal of motor, and red lead to (-) terminal. (The (-) terminal is indicated by arrow on the motor.)

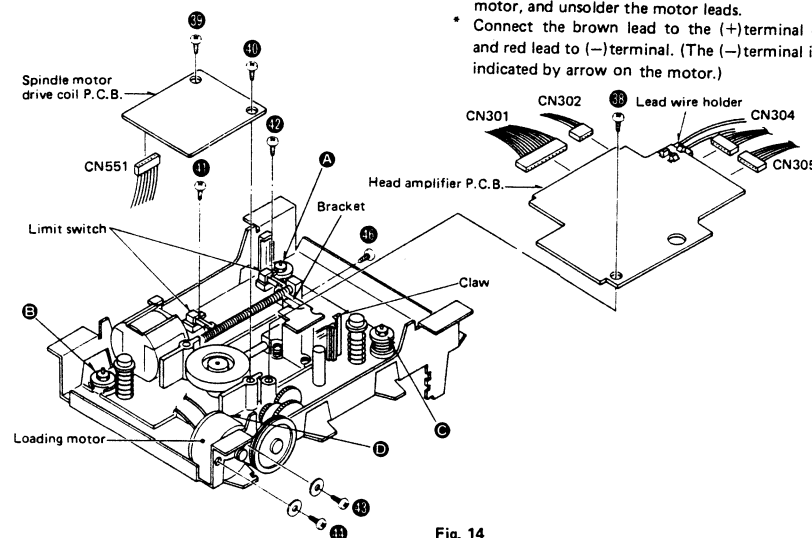


Fig. 14

### How to remove the turntable

1. Remove the spindle motor drive coil P.C.B. [Refer to "How to remove the head amplifier P.C.B. and spindle motor drive coil P.C.B."]
  2. Remove the C ring from the shaft of spindle motor. [Fig. 15]
  3. Remove the screw [Fig. 15 : ⑮] by use of a hexagonal wrench. Then the capstan shaft can be removed.
- \*If the turntable is removed, it is necessary to re-adjust the height of turntable.

### How to remove the laser pickup

1. Remove the head amplifier P.C.B. [Refer to "How to remove the head amplifier P.C.B."]
  2. Remove the bracket setscrew [Fig. 14 : ⑩] of pickup.
  3. Remove the loading guide holder setscrews [Fig. 11 : ② ~ ④], (Disc holder is fitted to loading guide.)
  4. Remove the guide rail retainer setscrew [Fig. 15 : ⑭], and pull the laser pickup out of the two guide rails.
  5. Pull out the 2 connectors [CN304, CN305] of head amplifier P.C.B. [Fig. 14]
- \*After replacing the pickup or fitting it again, it must be re-adjusted.

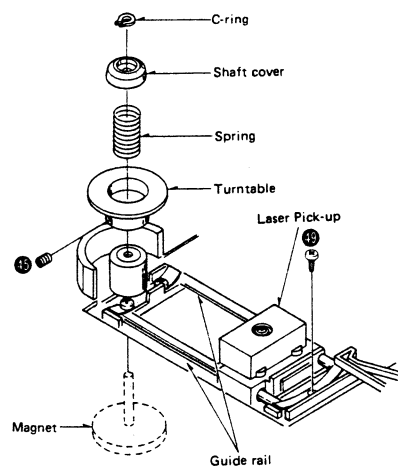


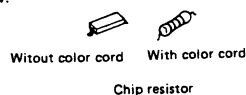
Fig. 15

## HOW TO REPLACE CHIPS

(Resistor, capacitor and jumper)

### Removing procedure

1. Completely remove the solder from both ends of the chip by use of solder sucker.
2. Touch the soldering iron to the end of the chip as shown in Fig. 16, then turn the tweezers in the direction of the arrow.



Do not re-use chip resistor or capacitor without color cord.

### Replacing procedure

1. Place solder on the foil where the chip is fitted. Then solder the chip by holding the soldering iron as shown in Fig. 17.

#### Note:

1. If the chip jumper is removed, connect a coated lead wire to the part. (See Fig. 18). Chip jumper is marked with "J" on the printed circuit board.

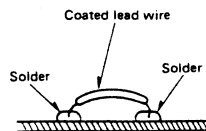


Fig. 18

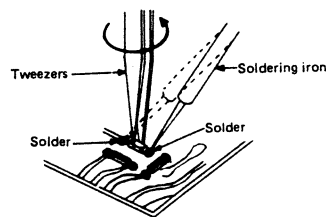


Fig. 16

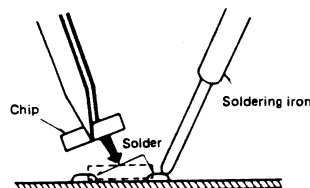


Fig. 17

### Note for replacing chips

1. Do not heat the chip more than 3 seconds.
2. Do not rub the electrode against the chip.
3. Use the tweezers with care not to damage the surface of the chip.
4. It is desirable to use a pencil type soldering iron. And use soldering iron less than 60W.

## HOW TO REPLACE IC'S (Small outline type)

| Replacing procedure |                                                                                                     | Cautions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                   | Reduce the amount of solder on each pin of the integrated circuit by use of a solder sucker.        | <ul style="list-style-type: none"> <li>● Recommended tool<br/>..... Special soldering iron (with spare chip) HM-354</li> <li>● Do not touch the soldering iron to the area for a long time. It may otherwise cause removal of the print foil.</li> <li>● When shifting the pin upward, do the job quickly while the solder is melting. If the solder is hard, it may cause removal or breakage of the print foil.</li> <li>● When using a pencil type soldering iron.                             <ol style="list-style-type: none"> <li>1. Completely remove the solder from each IC pin by use of solder sucker.</li> <li>2. Raise each pin by means of an eyelet, hold the pliers then remove IC package from P.C.B.</li> </ol> </li> </ul> |
| 2                   | Melt the solder on the pin (one electrode) with the soldering iron.                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3                   | While the solder is melting, shift the pin upward by the soldering iron to remove it from the foil. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4                   | Remove each pin from the foil according to the above-mentioned procedure.                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## RESISTOR AND CAPACITORS

- Notes:**
1. Part numbers are indicated on most mechanical parts. Please use this part number for parts orders.
  2. Important safety notice:  
Components identified by  $\Delta$  mark have special characteristics important for safety. When replacing any of these components use only manufacturer's specified parts.
  3. This "S" mark is service standard parts and may differ from production parts.

4. Unless otherwise specified.  
All resistors are in OHMS ( $\Omega$ ) K = 1000 $\Omega$ , M = 1000K $\Omega$ .  
All capacitors are in MICROFARADS ( $\mu$ F) P = 10<sup>-6</sup>  $\mu$ F.

### Numbering System of Resistor

Example

|      |         |       |           |       |
|------|---------|-------|-----------|-------|
| ERD  | 25      | F     | J         | 101   |
| Type | Wattage | Shape | Tolerance | Value |
| ERG  | 1       | AN    | J         | 2R2   |
| Type | Wattage | Shape | Tolerance | Value |

| Resistor Type     | Wattage   | Tolerance    |
|-------------------|-----------|--------------|
| ERD : Carbon      | 25 : 1/4W | F : $\pm$ 1% |
| ERG : Metal Oxide | 1 : 1W    | J : $\pm$ 5% |
| ERX : Metal Film  |           | G : $\pm$ 2% |

- ERD10TLJ  $\square\square\square$  → Chip type carbon (1/8W)  
 ECUV1H  $\square\square\square$  → Chip type ceramic  
 ERDS2TJ  $\square\square\square$  → Small type carbon (1/4W)  
 EROS2TKF  $\square\square\square$  → Small type metal film (1/4W)

### Numbering System of Capacitor

Example

|      |         |                 |           |             |
|------|---------|-----------------|-----------|-------------|
| ECKD | 1H      | 102             | Z         | F           |
| Type | Voltage | Value           | Tolerance | Peculiarity |
| ECEA | 50      | M               | R47       | R           |
| Type | Voltage | Peculiarity use | Value     | Special use |

| Capacitor Type      | Voltage   |              | Tolerance      |
|---------------------|-----------|--------------|----------------|
|                     | ECEA Type | Others       |                |
| ECEA : Electrolytic | 1A : 10V  | 1H : 50V DC  | J : $\pm$ 5%   |
| ECKD : Ceramic      | 1C : 16V  | 2H : 500V DC | K : $\pm$ 10%  |
| ECQM : Polyester    | 1E : 25V  | 1 : 100V     | Z : +80%, -20% |
| ECCD : Ceramic      | 1V : 35V  | KC : 400V AC | P : +100%, -0% |
| ECKF : Ceramic      | 1H : 50V  | HS : 400V AC | M : $\pm$ 20%  |
|                     | 1J : 63V  |              |                |
|                     | 50 : 50V  |              |                |